

IUG - Faculty of Engineering
Department of Computer Engineering
Data communications Lab
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Data Communications

Lab 1

Fundamentals of data communications and computer networks – part 1

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Data Communication

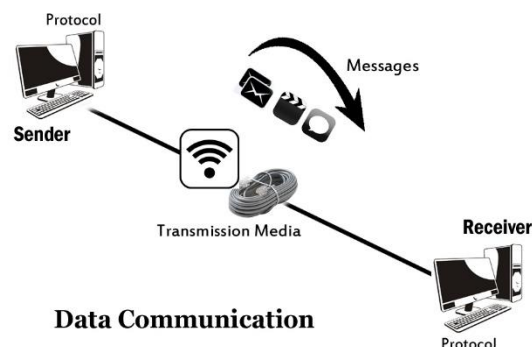
Data communications is the process of using computing and communication technologies to transfer data from one place to another, or between participating parties. It enables the movement of electronic or digital data between two or more network nodes, regardless of geographical location, technological medium or data contents.

Some common types of data communications technologies include telecommunications, computer networking and radio/satellite communication. Data communication usually requires the existence of a transportation or communication medium between the nodes wanting to communicate with each other, such as copper wire, fiber optic cables, or wireless signals.

For example, a common example of data communications is a computer connected to the Internet via a Wi-Fi connection, which uses a wireless medium to send and receive data from one or more remote servers.

The components of Data Communication System

- 1. Message** - It is the information to be communicated. Popular forms of information include text, pictures, audio, video etc. Text is converted to binary, number does not convert, image is converted to pixels, etc.
- 2. Sender** - It is the device which sends the data messages. It can be a computer, workstation, telephone handset etc.
- 3. Receiver** - It is the device which receives the data messages. It can be a computer, workstation, telephone handset etc.
- 4. Transmission Medium** - It is the physical path by which a message travels from sender to receiver. Some examples include twisted-pair wire, coaxial cable, radio waves etc.
- 5. Protocol** - It is a set of rules that need to be followed by the communicating parties in order to have successful and reliable data communication. You have already come across protocols such as Ethernet and HTTP.



Measuring Capacity of Communication Media

In data communication, the transmission medium is also known as channel. The capacity of a channel is the maximum amount of signals or traffic that a channel can carry. It is measured in terms of bandwidth and data transfer rate as described below:

Bandwidth

Bandwidth of a channel is the range of frequencies available for transmission of data through that channel. Higher the bandwidth, higher the data transfer rate. Normally, bandwidth is the difference of maximum and minimum frequency contained in the composite signals. Bandwidth is measured in Hertz (Hz).

$$1 \text{ KHz} = 1000 \text{ Hz}$$

$$1 \text{ MHz} = 1000 \text{ KHz} = 1000000 \text{ Hz}$$

Data Transfer Rate

Data travels in the form of signals over a channel. One signal carries one or more bits over the channel. Data transfer rate is the number of bits transmitted between source and destination in one second. It is also known as bit rate. It is measured in terms of bits per second (bps). The higher units for data transfer rates are:

$$1 \text{ Kbps} = 2^{10} \text{ bps} = 1024 \text{ bps}$$

$$1 \text{ Mbps} = 2^{20} \text{ bps} = 1024 \text{ Kbps}$$

$$1 \text{ Gbps} = 2^{30} \text{ bps} = 1024 \text{ Mbps}$$

$$1 \text{ Tbps} = 2^{40} \text{ bps} = 1024 \text{ Gbps}$$

Example

A user wants to upload a text document at the rate of 10 pages per 20 second. What will be the required data rate of the channel? (Assume that 1 page contains 1600 characters and each character is of 8 bits).

$$\begin{aligned} \textbf{Solution:} \text{ Required data rate } & \frac{(10 \times 1600 \times 8)}{20} \\ & = 6400 \text{ bps} = 6.25 \text{ Kbps} \end{aligned}$$

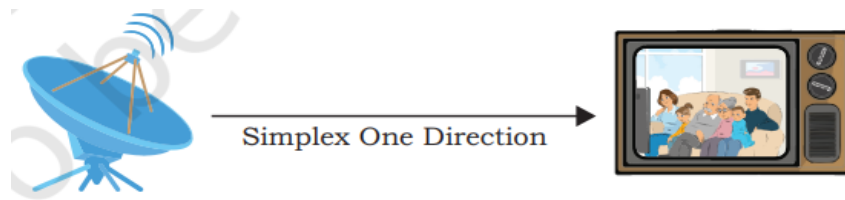
Data Flow techniques

Transmission mode refers to the mechanism of transferring of data between two devices connected over a network. It is also called Communication Mode. These modes direct the direction of flow of information. There are three types of transmission modes.

They are: **1. Simplex Mode** **2. Half duplex Mode** **3. Full duplex Mode**

1. SIMPLEX Mode

It is a one way or unidirectional communication between two devices in which one device is sender and other one is receiver. Devices use the entire capacity of the link to transmit the data. It is like a one-way street where vehicles can move in only one direction. For example, data entered through a keyboard or audio sent to a speaker are one-way communications.



2. HALF DUPLEX Mode

Half-duplex data transmission means that data can be transmitted in both directions on a signal carrier, but not at the same time. For example, on a local area network using a technology that has half-duplex transmission, one workstation can send data on the line and then immediately receive data on the line from the same direction in which data was just transmitted. Hence half- duplex transmission implies a bidirectional line (one that can carry data in both directions) but data can be sent in only one direction at a time. Example of half duplex is a walkie- talkie in which message is sent one at a time but messages are sent in both the directions.



3. FULL DUPLEX Mode

In full duplex system we can send data in both the directions as it is bidirectional at the same time in other words, data can be sent in both directions simultaneously. Example of Full Duplex is a Telephone Network in which there is communication between two persons by a telephone line, using which both can talk and listen at the same time.



Types of Connection in Computer Networks

A Network is nothing but a connection made through connection links between two or more devices. Devices can be a computer, printer or any other device that is capable to send and receive data. There are two ways to connect the devices:

1. Point-to-Point connection
2. Multipoint connection

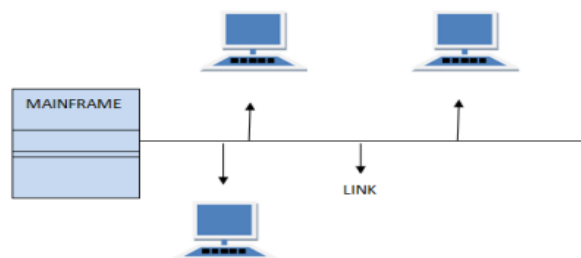
1. Point-To-Point Connection

Point-to-point connections provide a dedicated link between two devices. Each device can communicate only with those that are directly connected to it.



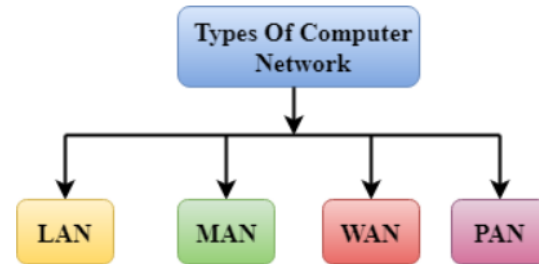
2. Multipoint Connection

In multipoint networks many devices share a single link or communication medium.



Types of computer Network:

Computer Network is a system of interconnected computers that enable the computers to communicate with each other and share their resources, data and applications.



PAN (Personal Area Network)

- Personal Area Network is a network arranged within an individual person, typically within a range of 10 meters.
- Personal Area Network is used for connecting the computer devices of personal use is known as **Personal Area Network**.
- Personal Area Network covers an area of 30 feet.
- Personal computer devices that are used to develop the personal area network are the laptop, mobile phones, media player and play stations.
- There are **two types** of Personal Area Network:
 - **Wireless Personal Area Network**: Wireless Personal Area Network is developed by simply using wireless technologies such as **WiFi**, **Bluetooth**. It is a low range network.
 - **Wired Personal Area Network**: Wired Personal Area Network is created by using the **USB**.

LAN (Local Area Network)

- Local Area Network is a group of computers connected to each other in a small area such as building, office.
- LAN is used for connecting two or more personal computers through a communication medium such as twisted pair, coaxial cable, etc.
- It is less costly as it is built with inexpensive hardware such as hubs, network adapters, and ethernet cables.
- The data is transferred at an extremely faster rate in Local Area Network.
- Local Area Network provides higher security.
- **Example of LAN**: home network

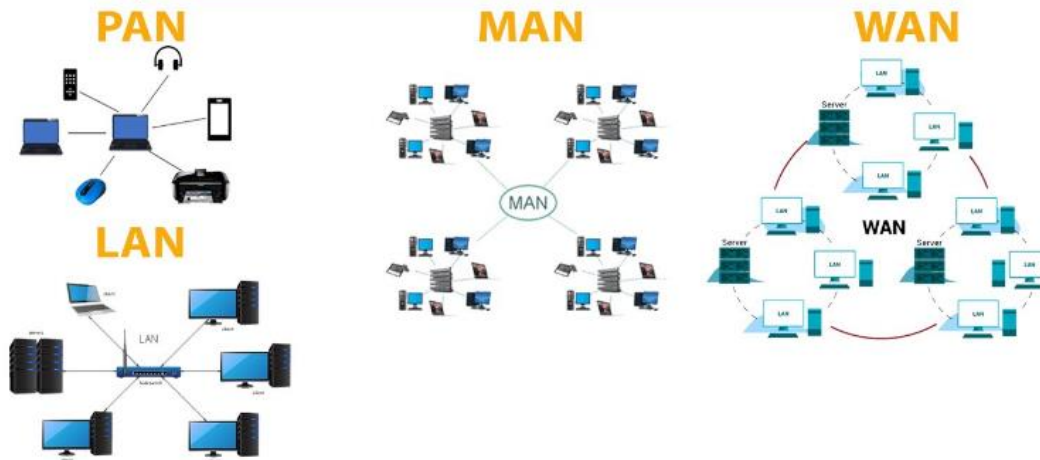
MAN (Metropolitan Area Network)

- A metropolitan area network is a network that covers a larger geographic area by interconnecting a different LAN to form a larger network.
- Government agencies use MAN to connect to the citizens and private industries.
- In MAN, various LANs are connected to each other through a telephone exchange line.
- It has a higher range than Local Area Network (LAN).

WAN (Wide Area Network)

- A Wide Area Network is a network that extends over a large geographical area such as states or countries.
- A Wide Area Network is quite bigger network than the LAN.
- A Wide Area Network is not limited to a single location, but it spans over a large geographical area through a telephone line, fibre optic cable or satellite links.
- The internet is one of the biggest WAN in the world.
- A Wide Area Network is widely used in the field of Business, government, and education.

Types of Computer Network



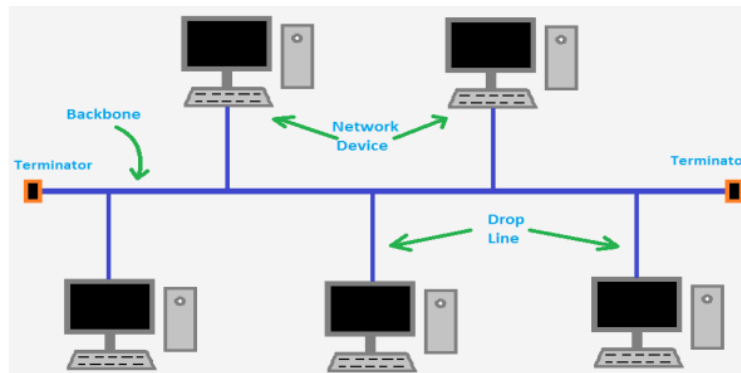
Network Topologies

Topology defines the structure of the network of how all the components are interconnected to each other. There are two types of topology: physical and logical topology.

Physical topology is the geometric representation of all the nodes in a network.

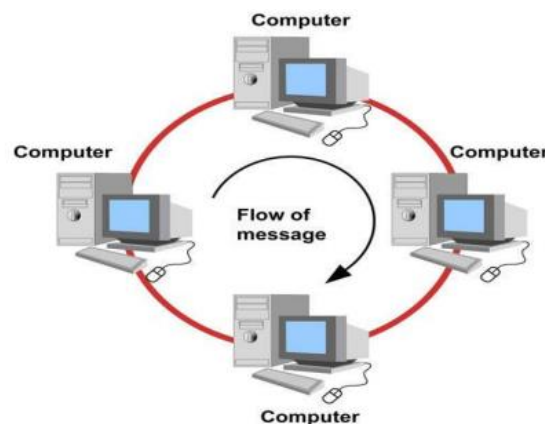
BUS Topology

- A bus topology consists of main System Bus with a terminator at each end in which all computers (node) are connected to the cable.
- A signal from the source travels in both directions to all computers connected on the bus cable until it finds the intended recipient. If the computer address does not match the intended address for the data, the computer ignores the data.



RING Topology

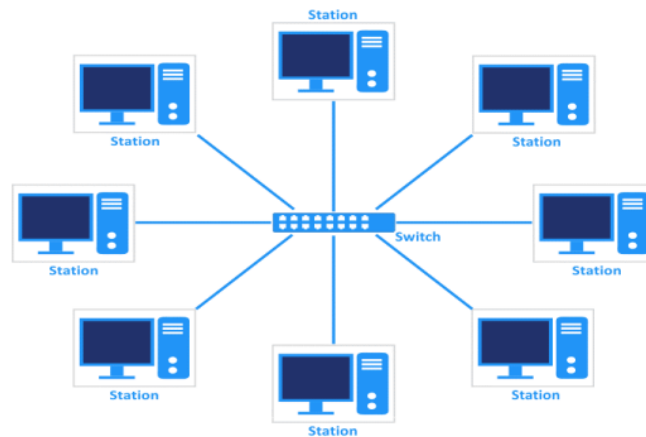
- In this all the computers (nodes) are connected to each-other in such a way that they make a closed loop.
- Each computer is connected to two other components on either side, and it communicates with these two adjacent neighbors. Data is sent around the ring until it reaches its final destination.



STAR Topology

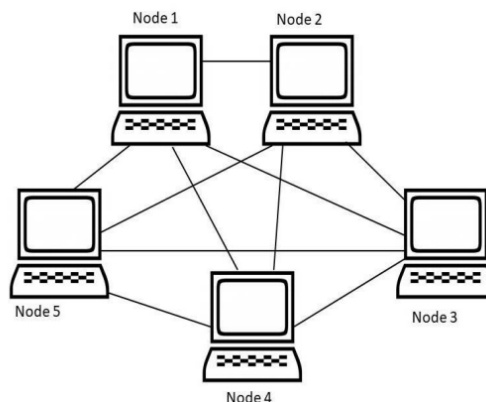
- In this every node (computer workstation or any other peripheral) is connected to central node or device called hub or switch)
- All the data on the star topology passes through the central device before reaching the intended destination.

- A hub forwards data packets to all the ports. A switch is more intelligent. It recognizes which computer is connected to which port based on the MAC address and saves this information in a table. When a switch receives a data packet, it determines the recipient and forwards the packet to the correct computer.
- An advantage of the star topology is the simplicity of adding additional nodes. The primary disadvantage of the star topology is that the hub/switch represents a single point of failure.
- Extended Star topology uses the star topology to be created. It links individual stars together by linking the hubs/switches. this will extend the length of the network.



MESH Topology

- This topology employs either or two schemes, called full mesh and partial mesh.
- In the full mesh topology, each computer is connected directly to each of the others. In the partial mesh topology, some computers are connected to all the others, and some are connected only to those other nodes with which they exchange the most data.



Hybrid Topology:

- The combination of various different topologies is known as Hybrid topology. A Hybrid topology is a connection between different links and nodes to transfer the data.
- When two or more different topologies are combined together is termed as Hybrid topology and if similar topologies are connected with each other will not result in Hybrid topology.
- For example, if there exist a ring topology in one branch of ICICI bank and bus topology in another branch of ICICI bank, connecting these two topologies will result in Hybrid topology.

